Procurement Fraud & Supplier Risk Analysis Dashboard

Project Report

Prepared By

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Tools Used

- PostgreSQL 17
- pgAdmin 4
- Python
- Psycopg2
- JSON Dataset
- SQL
- Power BI Desktop
- GitHub

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1. Executive Summary

Procurement fraud remains a significant challenge for many organizations due to supplier manipulation, invoice irregularities, duplicate vendors, unauthorized approvals, and weak procurement controls.

This project was developed to identify procurement fraud indicators and supplier risks using PostgreSQL, Python, SQL, and Power BI.

The solution follows a modern analytics architecture consisting of Raw, Staging, and Analytics layers. Procurement invoice data was extracted from a JSON source file, loaded into PostgreSQL, cleaned and transformed using SQL, and visualized through an interactive Power BI dashboard.

The dashboard enables business stakeholders to monitor supplier risks, identify high-risk invoices, analyze departmental spending patterns, and support data-driven procurement decisions.

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2. Business Problem

Procurement departments process large volumes of supplier invoices daily. Without proper monitoring, organizations may experience:

- Fraudulent supplier activities
- Duplicate supplier records
- Unauthorized approvals

- High-risk transactions
- Late payments
- Missing purchase order references
- Financial losses

Management requires a centralized reporting solution capable of highlighting procurement risks and supporting investigative activities.

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3. Project Objective

The objective of this project was to design and implement a procurement fraud and supplier risk monitoring solution capable of:

- Identifying high-risk suppliers
- Detecting potentially fraudulent invoices
- Monitoring procurement spending
- Highlighting approval irregularities
- Supporting procurement governance
- Providing interactive reporting for decision-makers

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4. Project Architecture

The project was designed using a three-layer data architecture commonly used in modern data engineering environments.

Raw Layer

Purpose:

Store original invoice records exactly as received from the source system.

Schema:

raw

Table:

raw.procurement_invoices

Staging Layer

Purpose:

Clean and standardize invoice data.

Activities:

- Remove inconsistencies
- Standardize text values
- Validate data types
- Prepare records for reporting

Schema:

staging

Table:

staging.procurement_invoices_clean

Analytics Layer

Purpose:

Create business-ready reporting views.

Schema:

analytics

Views:

- analytics.department_spend_summary
- analytics.supplier_risk_summary
- analytics.invoice_risk_flags

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5. Data Source

Dataset Type:

JSON File

File Name:

procurement_invoices_dirty.json

The dataset contains procurement invoice information including:

- Invoice ID
- Supplier Name
- Department
- Invoice Amount

- Invoice Date
- Payment Date
- Approval Status
- Supplier Country
- Purchase Order Reference

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6. Data Engineering Process

Step 1: Data Extraction

Python was used to read invoice data from a JSON file.

The data was extracted and prepared for loading into PostgreSQL.

Step 2: Database Connection

Python and Psycopg2 were used to establish a connection between the application and PostgreSQL.

Step 3: Data Loading

The extracted records were inserted into:

raw.procurement_invoices

Step 4: Data Cleaning

The staging layer was used to clean and standardize data.

Cleaning activities included:

- Standardizing supplier names
- Removing unnecessary spaces
- Formatting dates
- Validating invoice amounts
- Handling missing values

Step 5: Risk Calculation

Business rules were created to identify procurement risk indicators.

Risk indicators included:

High-Risk Supplier

Flagged when supplier risk conditions were met.

High-Value Invoice

Flagged when invoice amount exceeded defined thresholds.

Late Payment

Flagged when payment occurred after the due date.

Missing Purchase Order

Flagged when no purchase order reference existed.

Approval Risk

Flagged when approval controls were violated.

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7. Analytics Layer Design

Department Spend Summary

Purpose:

Measure spending by department.

Metrics:

- Total Spend
- Average Invoice Amount
- Total Invoices
- High-Risk Suppliers

Supplier Risk Summary

Purpose:

Measure supplier risk exposure.

Metrics:

- Total Invoices
- Total Invoice Value
- Supplier Risk Flags
- Total Risk Score

Invoice Risk Flags

Purpose:

Provide transaction-level risk monitoring.

Metrics:

- Invoice Risk Score
- High-Risk Supplier Flag
- High-Value Invoice Flag
- Missing PO Flag
- Approval Risk Flag
- Late Payment Flag

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8. Power BI Dashboard Development

Power BI was connected directly to PostgreSQL.

The analytics views were imported and used to create interactive visualizations.

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9. Dashboard Components

KPI Cards

Total Spend

£132.98K

Purpose:

Shows total procurement expenditure.

Total Invoices

10

Purpose:

Shows total invoices processed.

High-Risk Suppliers

3

Purpose:

Highlights suppliers requiring further review.

High-Value Invoices

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Purpose:

Highlights potentially risky high-value transactions.

Department Spend Analysis

Visual Type:

Horizontal Bar Chart

Purpose:

Compare spending across departments.

Supplier Risk Analysis

Visual Type:

Horizontal Bar Chart

Purpose:

Compare supplier risk scores.

Top Risky Invoices

Visual Type:

Bar Chart

Purpose:

Identify invoices with the highest risk scores.

Invoice Risk Flag Analysis

Visual Type:

Table

Purpose:

Display detailed invoice-level risk indicators.

Department Slicer

Visual Type:

Interactive Filter

Purpose:

Allow users to filter dashboard results by department.

Options:

- Clinical
- Finance
- IT
- Operations
- Procurement

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10. Key Insights

Insight 1: Procurement Department Spending

The Procurement department recorded the highest overall spending compared to other departments.

Business Impact:

Higher spending departments require stronger procurement controls due to increased fraud exposure.

Insight 2: High-Risk Suppliers

Three suppliers accumulated the highest risk scores.

Business Impact:

These suppliers should be prioritized for further investigation and supplier due diligence.

Insight 3: High-Risk Invoices

Only a small number of invoices generated the highest risk scores.

Business Impact:

Auditors can focus investigations on a limited set of high-risk transactions.

Insight 4: Supplier Naming Inconsistencies

The project identified supplier naming variations such as:

- Alpha Medical Ltd
- Alpha Medical Limited

and

- Delta Facilities Ltd
- Delta Facilities Limited

Business Impact:

Supplier master-data inconsistencies may cause inaccurate reporting and duplicate supplier records.

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11. Recommendations

Recommendation 1

Implement supplier master-data governance.

Supplier names should be standardized before reporting.

Recommendation 2

Introduce automated fraud alerts.

Invoices exceeding risk thresholds should automatically trigger investigations.

Recommendation 3

Conduct quarterly supplier risk reviews.

Focus on suppliers with the highest cumulative risk scores.

Recommendation 4

Implement secondary approval controls.

High-risk invoices should require additional authorization.

Recommendation 5

Improve procurement data quality processes.

Regular validation checks should be performed on supplier records.

12. Business Value

This solution provides several business benefits:

- Improved fraud detection
- Better supplier monitoring
- Enhanced procurement governance
- Reduced financial risk
- Improved audit readiness
- Faster decision-making
- Increased visibility into procurement operations

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13. Future Enhancements

Potential future improvements include:

- Automated email alerts
- Real-time dashboard refresh
- Machine learning fraud detection models
- Supplier risk prediction
- Integration with ERP systems
- Azure Data Factory pipelines
- Azure SQL Database deployment
- Cloud-based reporting architecture

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14. Conclusion

This project successfully demonstrated an end-to-end procurement fraud and supplier risk analytics solution using PostgreSQL, Python, SQL, and Power BI.

The implementation provided stakeholders with visibility into procurement spending, supplier risks, invoice irregularities, and operational controls through an interactive dashboard.

The solution demonstrates practical skills in data engineering, data analysis, business intelligence, SQL development, data modeling, and dashboard design, making it suitable for Data Analyst, BI Analyst, and Junior Data Engineer portfolio presentations.